INST327 - Database Design and Modeling

Section 0203

Diabetes Patient Analysis Dataset

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Group G3

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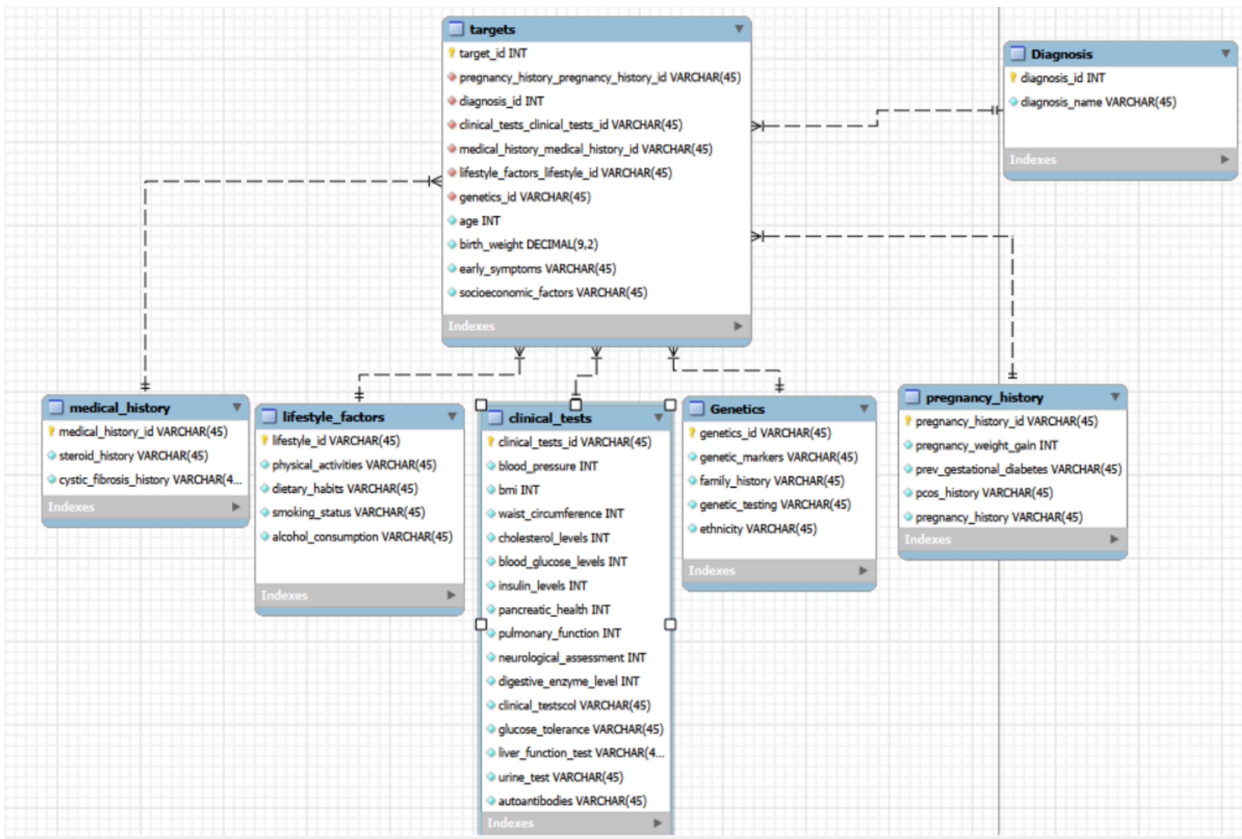
Project Progress Proposal

Database ER Diagram:



Database Description:

Our database includes a group of unique targets that each have their own individual attributes that are crucial in research for each target's correlating diagnoses. The dataset generally includes rows of individuals (targets) and their attributes that relate to their diabetes diagnosis. The database includes an entity that includes their actual diagnosis which can range from type 1 diabetes, LADA, gestational diabetes, steroid-induced diabetes, etc. Related attributes of the lifestyle factors table of the target that attains to diabetes include physical activity, dietary healthiness, smoking and drinking activity,, and more. There is also a genetics entity containing genetic markers, autoantibodies, family history with diabetes, genetic testing, etc. Out of these entities, the clinical tests table is the most important for comprehensive diabetes research. Some attributes within this theme include insulin levels, BMI, blood pressure, pulmonary function, digestive enzyme levels, urine test results, etc. which are all significant tests for individuals diagnosed with diabetes. We decided to keep all attributes from the original dataset since they all heavily contribute to the causation and are significant to different fields of research for our selected scope of diabetes. Each entity in the database also contains its own primary key (medical_tests_id, genetics_id, etc.), and the targets table has each table's primary key in it. This is due to the target entity being a child table to every other table as attributes from each table can have many instances in the targets table. For example, a row in the pregnancy_history table can show up many times in the targets table. The same applies to the rest of the entities.

Sample Data:

For our new proposed sample data, our data will only include some of the more common types of diabetes and not the rare ones. This would include type 1 and 2 diabetes, gestational diabetes, LADA, steroid-induced diabetes, prediabetes, and cystic fibrosis-related diabetes. After reducing the dataset down to only these diagnoses, we decided to then reduce the dataset down to 100 targets due to the original dataset having 70,000 targets. We did this for not only this project's requirements, but it is also for faster processing and better quality while still maintaining the dataset's essential information.

Progress Report:

With this report, our team has gone through two extensive meetings in order to discuss our sample data, how our ER diagram will be represented, the overall structure of our database, and our feedback on our proposal. The feedback stated that medical test results and general test results tables should be combined due to the attributes containing similar data. There was also feedback on switching socioeconomic factors and ethnicity into the targets table. Another piece of feedback recommended us to filter and narrow down the scope instead of selecting the first 10,000. With this feedback we actually decided to still reduce the data to 100 targets initially to hit the qualification for the project as we were authorized to do so. We also decided not to switch ethnicity to targets because the ethnicity attribute doesn't have data on the target's actual ethnicity, but if their ethnicity is generally at high or low risk of diabetes. So we concluded that ethnicity belongs within the genetics table. Although, we did actually put socioeconomic factors into the targets table because it pertains to each individual target.

Changes from the Initial Proposal:

We initially planned to analyze the whole dataset, which included all of the various forms of diabetes that were covered in the file, but as we met with our mentor we found it more helpful to select only a few forms of diabetes instead along with decreasing the total number of sample data as a whole to 100 people. This way we can dive deeper into the role that differences in lifestyle choices, habits, and genetics have on select forms of diabetes. We are now only focusing on the most common forms of diabetes to reduce the dataset in order to gain a better insight. We also changed the relationships from our entities in our database. After extensive questions and research, we've changed every entity to have a one to many relationship with the targets table with their own primary key.

By restructuring our data this way, we will have information that is working with each other consistently, rather than thousands of different one to one relationships that we can't make any conclusions on. Along with this, we refined the relationships to have better queries which leads to more clear results that we can draw from. In essence, the changes we made laid a new foundation for our project. We now have a much cleaner picture for what our goal actually is. Now we can focus more on patterns that have meaning and can accurately make conclusions, rather than piecing things together from chunks of thousands of contrasting data.

Diversity, Equity, and Inclusion Considerations:

The revised dataset includes helpful variables like age, socioeconomic status, lifestyle habits, and pregnancy-related information, which make it possible to compare different groups and see how environmental factors influence diabetes, however it still leaves out some important considerations.

A major problem is that the dataset only includes people who can get pregnant because it includes pregnancy-related traits. This means that a large part of the diabetic population is not included. This limitation reduces the dataset's inclusiveness and constrains the range of research inquiries that can be explored concerning diabetes in various populations. Other crucial information like sex, whether or not they have a disability, and whether or not they have access to healthcare is still missing. Without this information, it would be more difficult to determine how different social factors affect individuals with diabetes.

Another problem with this data set is how it represents ethnicity - it categorizes people under “high risk” or “low risk.” By doing this, it essentially categorizes race as only a biological difference rather than considering how environmental factors affect different people.

Data Privacy, Fair Use, Other Ethical Considerations:

The information in this dataset has been verified to not show Protected Health Information . Fields such as genetic markers, pregnancy information, and test results, cease to be protected under the Health Insurance Portability and Accountability Act (HIPAA) once they have their names, dates, special test codes, geographic information, and any of the other standard 18 identifiers for PHI stripped under the Safe Harbor method (UC Berkeley, 2025). Furthermore, **Age** would be the only identifying factor: IF there are entries with patients 90 and over, which does not ring true for this dataset (Office for Civil Rights, 2025). The data should also be considered of fair use, as the data when inspected on Kaggle, seems to have perfect mathematical distributions between almost every single one of its columns, which can be used to indicate synthetic data not attributed to any patient records (Batra, 2024).

In accordance with our due diligence, we will continue to navigate and make analysis through our database to determine if the data is synthetic or genuine patient data. As this is a database project to showcase our teamwork and skills and will only be used in educational environments, we do not need to decide ethical considerations regarding its use in healthcare (Office for Civil Rights, 2025).

Plans for Remaining Work:

After finalizing our database, we’ll implement the schema and make sure that all of the relationships and tables are correct. Along with this we are going to import our edited dataset with 100 samples and remove any unnecessary information. Once this part is finalized as well, we can create queries in SQL to make connections between forms of diabetes and contributing factors. We will also test our queries and make sure that they return results that we can actually work with. With that, we are planning to analyze the outputs to recognize patterns and trends that different diabetes has with different lifestyle choices, habits, and genetics. We are also going to start to compile everything that we have found into the framework of our final report.

Citations

Batra, A. (2024). *Diabetes Dataset*. Kaggle.com. https://www.kaggle.com/datasets/ankitbatra1210/diabetes-dataset/data

Office for Civil Rights. (2025, February 3). *Methods for De-identification of PHI*. HHS.gov. https://www.hhs.gov/hipaa/for-professionals/special-topics/de-identification/index.html

UC Berkeley. (2025). *HIPAA PHI: Definition of PHI and list of 18 identifiers*. Berkeley.edu. https://cphs.berkeley.edu/hipaa/hipaa18.html